

IS 3063 Term Project — Deliverable 3

Google Ads Agency Database — Alamo Digital Marketing

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Note: I am completing this project individually (approved by the instructor), so this document serves as the group proposal and the attached ERD is both the individual and group ERD.

Instructor Feedback on Deliverables 1 and 2 and Changes Made

Feedback received: The Deliverable 1 ERD file was not created using the required data model template in Microsoft Visio (the diagram was submitted in a generic format).

Change made in response: The ERD has been completely redrawn in desktop Microsoft Visio using the Crow's Foot Database Notation data model template, with primary keys, foreign keys, crow's-foot cardinality symbols, identifying (solid) vs. non-identifying (dashed) relationships, and the supertype/subtype constraint properly specified. The corrected Visio file is included with this deliverable.

Feedback received (Deliverable 2): The 1:M crow's-foot notation between the CLIENT and CONTRACT entities should be reversed; the foreign key should always be on the many side of a 1:M relationship.

Change made in response: The Client-Contract relationship has been redrawn in the ERD with the "one and only one" crow's-foot symbol on both ends, removing the crow's foot that had been placed on the CLIENT side. This relationship is intentionally the 1:1 relationship required for this project: each client signs exactly one contract and each contract belongs to exactly one client, as stated in business rule (a) in Section 3. The foreign key (ClientID) is correctly located in the CONTRACT entity, and the one-to-one relationship is enforced in the Access database by indexing CONTRACT.ClientID as "Yes (No Duplicates)," which prevents any client from being linked to a second contract. Microsoft Access confirms the relationship type as One-To-One with referential integrity enforced.

1. Description of the business context and related data management problem(s)

Alamo Digital Marketing is a small digital marketing agency that manages Google Ads campaigns on behalf of local businesses such as restaurants, law firms, and home-service companies. For each client, the agency's staff builds and maintains campaigns inside the Google Ads platform: each campaign contains ad groups, each ad group contains ads and bids on a set of keywords, and performance must be monitored continuously against the client's budget.

While Google Ads stores the advertising data itself, everything about running the agency as a business lives outside of it. Client contact information, signed contracts, monthly retainer fees, employee-to-account assignments, and invoices are currently scattered across spreadsheets, email threads, and individual employees' notes. This fragmentation creates several recurring data management problems: management cannot quickly answer which employees are responsible for which client accounts or whether any employee is overloaded; unpaid invoices are discovered late because billing records are not linked to client records; there is no unified view connecting a client's contract terms to the campaigns

and budgets being run for them; and when a remarketing campaign is created from an earlier campaign, the connection between the two is not recorded anywhere, making performance comparisons impossible.

This project builds a database application that integrates the agency's client management, campaign structure, staffing assignments, and billing into a single system. The database mirrors the real hierarchy of Google Ads (Client → Campaign → Ad Group → Ads and Keywords) while adding the business-side records (contracts, assignments, invoices) that Google Ads does not track, giving management one place to monitor operations, workloads, and revenue.

2. The entities and the attributes (with description, constraint and data types)

Entity: EMPLOYEE (supertype) — Every person employed by the agency. Subtypes: Account Manager, Ad Specialist, Analyst (overlapping, partial specialization).

Attribute	Description	Constraints	Data Type
<u>EmployeeID</u>	6-digit ID of each employee	PK	CHAR(6)
FirstName	Employee's first name	Not Null	VARCHAR2(30)
LastName	Employee's last name	Not Null	VARCHAR2(30)
Email	Company email address	Unique, Not Null	VARCHAR2(60)
HireDate	Date the employee was hired	Not Null	DATE
SupervisorID	EmployeeID of this employee's supervisor (unary relationship)	FK (EMPLOYEE)	CHAR(6)

Entity: ACCOUNT MANAGER (subtype of EMPLOYEE) — Employees who own client relationships.

Attribute	Description	Constraints	Data Type
<u>EmployeeID</u>	ID of the employee (shared with supertype)	PK, FK	CHAR(6)
ClientCapacity	Maximum number of clients this manager can handle	Not Null, >0	INTEGER

Entity: AD SPECIALIST (subtype of EMPLOYEE) — Employees who build and optimize campaigns.

Attribute	Description	Constraints	Data Type
<u>EmployeeID</u>	ID of the employee (shared with supertype)	PK, FK	CHAR(6)
GoogleCertID	Google Ads certification number	Unique	VARCHAR2(20)
CertExpiration	Date the certification expires		DATE

Entity: ANALYST (subtype of EMPLOYEE) — Employees who prepare performance reporting.

Attribute	Description	Constraints	Data Type
<u>EmployeeID</u>	ID of the employee (shared with supertype)	PK, FK	CHAR(6)

PrimaryTool	Main reporting tool used (e.g., Looker Studio)		VARCHAR2(30)
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Entity: CLIENT — A business that hires the agency to manage its Google Ads.

Attribute	Description	Constraints	Data Type
<u>ClientID</u>	6-digit ID of each client	PK	CHAR(6)
CompanyName	Legal name of the client business	Not Null	VARCHAR2(60)
Industry	Client's industry (e.g., restaurant, law firm)		VARCHAR2(40)
ContactName	Primary contact person at the client	Not Null	VARCHAR2(50)
ContactEmail	Primary contact's email	Not Null	VARCHAR2(60)
StartDate	Date the client relationship began	Not Null	DATE

Entity: CONTRACT — The signed service agreement for a client (1:1 with CLIENT).

Attribute	Description	Constraints	Data Type
<u>ContractID</u>	8-digit ID of each contract	PK	CHAR(8)
ClientID	Client this contract belongs to	FK, Unique, Not Null	CHAR(6)
SignedDate	Date the contract was signed	Not Null	DATE
EndDate	Date the contract term ends	> SignedDate	DATE
MonthlyFee	Monthly management fee	Not Null, >0	DECIMAL(8,2)
Status	Active, Expired, or Terminated	Default: Active	VARCHAR2(12)

Entity: CAMPAIGN — A Google Ads campaign run for a client.

Attribute	Description	Constraints	Data Type
<u>CampaignID</u>	10-digit ID of each campaign	PK	CHAR(10)
ClientID	Client this campaign belongs to	FK, Not Null	CHAR(6)
CampaignName	Name of the campaign	Not Null	VARCHAR2(60)
Objective	Campaign goal (Sales, Leads, Traffic, Awareness)	Not Null	VARCHAR2(20)
DailyBudget	Daily spending limit in dollars	Not Null, >0	DECIMAL(7,2)
StartDate	Date the campaign starts	Not Null	DATE
Status	Enabled, Paused, or Ended	Default: Enabled	VARCHAR2(10)
SourceCampaignID	Campaign this remarketing campaign is based on (unary relationship)	FK (CAMPAIGN)	CHAR(10)

Entity: AD GROUP — A themed group of ads and keywords within a campaign.

Attribute	Description	Constraints	Data Type
<u>AdGroupID</u>	10-digit ID of each ad group	PK	CHAR(10)

CampaignID	Campaign this ad group belongs to	FK, Not Null	CHAR(10)
AdGroupName	Name of the ad group	Not Null	VARCHAR2(60)
DefaultBid	Default maximum cost-per-click bid	>0	DECIMAL(5,2)
Status	Enabled or Paused	Default: Enabled	VARCHAR2(10)

Entity: AD — An individual advertisement within an ad group.

Attribute	Description	Constraints	Data Type
<u>AdID</u>	10-digit ID of each ad	PK	CHAR(10)
AdGroupID	Ad group this ad belongs to	FK, Not Null	CHAR(10)
AdType	Responsive Search, Display, or Video	Not Null	VARCHAR2(20)
Headline	Primary headline text	Not Null	VARCHAR2(30)
DescriptionText	Ad description text	Not Null	VARCHAR2(90)
FinalURL	Landing page the ad links to	Not Null	VARCHAR2(120)
Status	Enabled, Paused, or Disapproved	Default: Enabled	VARCHAR2(12)

Entity: KEYWORD — A search term the agency can bid on for any client.

Attribute	Description	Constraints	Data Type
<u>KeywordID</u>	8-digit ID of each keyword	PK	CHAR(8)
KeywordText	The search phrase itself	Unique, Not Null	VARCHAR2(80)
AvgSearchVolume	Average monthly searches	>=0	INTEGER

Entity: AD GROUP KEYWORD (associative entity between AD GROUP and KEYWORD) — Records each keyword added to each ad group, with its bid settings.

Attribute	Description	Constraints	Data Type
<u>AdGroupID</u>	Ad group using this keyword	PK, FK1	CHAR(10)
<u>KeywordID</u>	Keyword being used	PK, FK2	CHAR(8)
MatchType	Broad, Phrase, or Exact	Not Null	VARCHAR2(10)
MaxCPCBid	Maximum cost-per-click bid for this keyword in this ad group	>0	DECIMAL(5,2)
QualityScore	Google's 1-10 quality rating	Between 1 and 10	INTEGER
DateAdded	Date the keyword was added to the ad group	Not Null	DATE

Entity: ACCOUNT ASSIGNMENT (associative entity between EMPLOYEE and CLIENT) — Records which employees work on which client accounts.

Attribute	Description	Constraints	Data Type
<u>EmployeeID</u>	Employee assigned to the account	PK, FK1	CHAR(6)
<u>ClientID</u>	Client account being worked on	PK, FK2	CHAR(6)

RoleOnAccount	Employee's role for this client (Lead, Support)	Not Null	VARCHAR2(15)
AssignedDate	Date the assignment began	Not Null	DATE
HoursPerWeek	Weekly hours budgeted to this account	>0	DECIMAL(4,1)

Entity: INVOICE — A monthly bill issued to a client.

Attribute	Description	Constraints	Data Type
<u>InvoiceID</u>	10-digit ID of each invoice	PK	CHAR(10)
ClientID	Client being billed	FK, Not Null	CHAR(6)
InvoiceDate	Date the invoice was issued	Not Null	DATE
BillingPeriod	Month and year covered (e.g., 2026-07)	Not Null	CHAR(7)
AmountDue	Total amount billed	Not Null, >0	DECIMAL(9,2)
Status	Unpaid, Paid, or Overdue	Default: Unpaid	VARCHAR2(10)
PaidDate	Date payment was received		DATE

3. Business rules that define all the relationships among entities, the constraints, and the attribute domains

Each relationship below is defined by two bi-directional business rules specifying minimum and maximum cardinalities.

a. CLIENT – CONTRACT (1:1)

- Each client must have one and only one contract (min 1, max 1).
- Each contract must belong to one and only one client (min 1, max 1).

b. CLIENT – CAMPAIGN (1:M)

- A client may have zero or many campaigns (min 0, max many).
- Each campaign must belong to one and only one client (min 1, max 1).

c. CAMPAIGN – AD GROUP (1:M)

- A campaign must contain at least one and possibly many ad groups (min 1, max many).
- Each ad group must belong to one and only one campaign (min 1, max 1).

d. AD GROUP – AD (1:M)

- An ad group may contain zero or many ads (min 0, max many).
- Each ad must belong to one and only one ad group (min 1, max 1).

e. AD GROUP – AD GROUP KEYWORD (1:M, part of the M:N between AD GROUP and KEYWORD)

- An ad group may have zero or many keyword entries (min 0, max many).
- Each ad group keyword entry must belong to one and only one ad group (min 1, max 1).

f. KEYWORD – AD GROUP KEYWORD (1:M, part of the M:N between AD GROUP and KEYWORD)

- A keyword may appear in zero or many ad group keyword entries (min 0, max many).
- Each ad group keyword entry must reference one and only one keyword (min 1, max 1).

g. EMPLOYEE – ACCOUNT ASSIGNMENT (1:M, part of the M:N between EMPLOYEE and CLIENT)

- An employee may hold zero or many account assignments (min 0, max many).
- Each account assignment must belong to one and only one employee (min 1, max 1).

h. CLIENT – ACCOUNT ASSIGNMENT (1:M, part of the M:N between EMPLOYEE and CLIENT)

- A client must have at least one and possibly many account assignments (min 1, max many).
- Each account assignment must reference one and only one client (min 1, max 1).

i. CLIENT – INVOICE (1:M)

- A client may receive zero or many invoices (min 0, max many).
- Each invoice must be issued to one and only one client (min 1, max 1).

j. EMPLOYEE – EMPLOYEE (unary, supervises, 1:M)

- An employee may supervise zero or many other employees (min 0, max many).
- An employee may be supervised by at most one other employee (min 0, max 1).

k. CAMPAIGN – CAMPAIGN (unary, remarketing-of, 1:M)

- A campaign may serve as the source for zero or many remarketing campaigns (min 0, max many).
- A remarketing campaign may be based on at most one source campaign (min 0, max 1).

l. EMPLOYEE supertype – ACCOUNT MANAGER / AD SPECIALIST / ANALYST subtypes

- Each employee may belong to zero, one, or multiple subtypes (partial specialization; overlapping constraint — one employee can be both an Ad Specialist and an Analyst at a small agency).
- Each Account Manager, Ad Specialist, or Analyst record must correspond to one and only one employee (min 1, max 1).

Additional constraints: A keyword cannot appear more than once within the same ad group (enforced by the composite primary key of AD GROUP KEYWORD). An employee cannot supervise himself or herself. A campaign cannot be a remarketing campaign of itself. QualityScore must be an integer between 1 and 10. MatchType is restricted to the domain {Broad, Phrase, Exact}. Campaign Objective is restricted to the domain {Sales, Leads, Traffic, Awareness}. Invoice Status is restricted to the domain {Unpaid, Paid, Overdue}.

4. Updated ERD

The ERD is provided in the attached Visio file (Deliverable_2_LastName.vsd), drawn in desktop Microsoft Visio using the Crow's Foot Database Notation data model template. No structural changes were made to the ERD since Deliverable 1; the diagram was redrawn in the required template format.

Relationship checklist: 1:1 (Client–Contract), 1:M (Client–Campaign, Campaign–Ad Group, Ad Group–Ad, Client–Invoice), M:N broken into 1:M through associative entities (Ad Group–Keyword via AD GROUP KEYWORD; Employee–Client via ACCOUNT ASSIGNMENT), unary (Employee supervises Employee;

Campaign remarketing-of Campaign), and supertype/subtype (Employee → Account Manager, Ad Specialist, Analyst; overlapping, partial). Total entities: 13 (the supertype hierarchy counts as one).

5. Summary of the functionality of the application

The application allows agency staff and management to register clients and contracts, build and view the full campaign structure for any client, assign employees to client accounts, monitor workloads, and generate and track monthly invoices.

10 specific functionalities implemented as queries in Access in Deliverable 3. The requirement number(s) each functionality satisfies are indicated in parentheses at the end of each item.

F1. Client contract overview: Displayed is each client's company name, industry, contract monthly fee, contract status, and contract end date, so management can see every client's terms in one list. Entities involved: CLIENT, CONTRACT. (#2)

F2. Unpaid billing by client: For each client, displayed is the total amount (Sum) of invoices whose status is Unpaid or Overdue, grouped by client, so overdue accounts are caught early. Entities involved: CLIENT, INVOICE. (#2, #4, #5)

F3. Average campaign budget per client: Displayed is the average (Avg) daily budget of campaigns for each client, grouped by client, so budget levels can be compared across accounts. Entities involved: CLIENT, CAMPAIGN. (#2, #4, #5)

F4. Keyword health by ad group: For each ad group, count (Count) the number of keywords attached to it and display the average (Avg) quality score of those keywords, so low-quality ad groups can be identified. Entities involved: AD GROUP, AD GROUP KEYWORD, KEYWORD. (#2, #3, #4, #5, #6)

F5. Highest bid in each campaign: For each campaign, display the maximum (Max) MaxCPCBid among all keywords in all of its ad groups, so management can monitor bidding levels. Entities involved: CAMPAIGN, AD GROUP, AD GROUP KEYWORD, KEYWORD. (#2, #3, #4)

F6. Employee supervision report: List each employee's name together with the name of his or her supervisor, a self-join will be used of the EMPLOYEE table on SupervisorID. Entities involved: EMPLOYEE (self-join). (#7)

F7. Certification expiration alert: we will list each Ad Specialist's first and last name (from the EMPLOYEE supertype) with the certification ID and expiration date (from the AD SPECIALIST subtype), so expiring Google Ads certifications are flagged. Entities involved: EMPLOYEE, AD SPECIALIST. (#2, #8)

F8. Employee workload: For each employee, we will use (Count) to count the number of client accounts assigned to that employee through Account Assignment, grouped by employee, so overloaded staff can be identified. Entities involved: EMPLOYEE, ACCOUNT ASSIGNMENT, CLIENT. (#2, #3, #4, #5, #6)

F9. Full ad listing for a campaign: For a selected campaign, we will list every ad (headline, type, status) in every ad group of that campaign, giving the account team a complete creative inventory. Entities involved: CAMPAIGN, AD GROUP, AD. (#2, #3)

F10. Remarketing lineage: We will list each remarketing campaign together with the name of the source campaign it is based on, using a self-join of the CAMPAIGN table on SourceCampaignID. Entities involved: CAMPAIGN (self-join). (#7)